

Nikunj Pradhan

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EDUCATION

Missouri University of Science and Technology

Masters of Computer Science

Bachelor of Science: Computer Science

GPA: 4.0/4.0

Aug 2025 - May 2027

Aug 2023 - Dec 2026

EXPERIENCE

Data Science Intern

May 2026 – Aug 2026

Quest Analytics

Overland Park, KS

- Architected a Medallion ETL pipeline on Databricks to ingest 20+ state data sources, standardizing 12M+ records into a deduplicated 3M-provider gold dataset
- Expanded internal Medicaid provider coverage by 40% by identifying 500K+ previously absent providers, creating a ground-truth dataset consumed by downstream product and analytics teams
- Built a data preprocessing pipeline using fuzzy matching and semantic embeddings to normalize schemas, standardize addresses, and maintain UUID-based lineage across 20+ heterogeneous sources
- Resolved 3,000+ provider records by building a LangGraph-based research tool for entity resolution, automating iterative search, extraction, and validation while routing tasks across LLMs to optimize inference cost and latency.

ML Software Engineer Co-Op

Jan 2025 – Aug 2025

Hunter Engineering Company

Bridgeton, MO

- Delivered a containerized FastAPI inference service with sub-750ms latency for production testing.
- Engineered ELT pipelines to batch, ingest, and process 2M+ images from 600+ worksites into AWS S3 and SQL Server
- Optimized end-to-end OCR license plate recognition pipeline, boosting classification accuracy from 68% to 84% across internal degraded image quality and imbalanced datasets
- Built ML infrastructure tools including model performance dashboards, real-time dataset analytics platforms, and AI-assisted annotation services.
- Accelerated distributed multi-GPU training workflows by 66% through close collaboration with R&D teams.

Software Engineer

June 2025 – July 2025

Tally Receipts

St. Louis, MO

- Migrated 9 MongoDB collections to AWS DynamoDB, redesigning schemas and rewriting 7 APIs to support the new database architecture.
- Built an event-driven card-linking pipeline API Gateway, SQS, and DynamoDB; implemented signature verification, event-level idempotency, sensitive-field redaction, and automated card status updates from webhook events.
- Engineered an LLM-powered receipt processing workflow with Amazon Bedrock that extracted structured transaction data and matched receipts to relevant calendar events, integrating Google and Outlook Calendar.

PROJECTS

Devanagari OCR Pipeline Application | *FastAPI, Python, Pytorch, Docker, ONNX*

- Optimized CPU OCR inference with ONNX Runtime, processing batches of 16 document images in 5s via a Dockerized FastAPI service; developed a custom CRNN-based OCR pipeline achieving 7.1% CER and 16.9% WER.

Movie Recommendation & Watchlist Platform | *Spring Boot, FastAPI, React, PostgreSQL, Redis, Kubernetes*

- Developed and deployed a full-stack application on Kubernetes, using Spring Boot for core backend services, FastAPI for ML recommendation inference, and Redis to cache authentication sessions, user events, and watchlist data.

Real-Time BlackJack Statistical Engine | *FastAPI, Apache Kafka, Numpy*

- Architected a high-throughput CV and Apache Kafka pipeline feeding a NumPy-vectorized Monte Carlo engine, simulating 10^5 states to optimize Expected Value (EV) for Blackjack outcomes.

TECHNICAL SKILLS

Languages: Python, Java, C++, SQL, JavaScript, HTML/CSS

Frameworks & APIs: FastAPI, Spring Boot, React, Node.js, REST APIs

Distributed Systems: Apache Kafka, Apache Spark (PySpark), Redis, WebSockets

Databases: PostgreSQL, MongoDB, SQL Server, DynamoDB, pgvector, MinIO, Amazon S3

Cloud & DevOps: AWS, Docker, Kubernetes, Git, CI/CD, Linux, Databricks, GitHub Actions

Machine Learning: PyTorch, Scikit-learn, XGBoost, LangChain, RAG, Transformers

Libraries & Tools: Pandas, NumPy, OpenCV, Playwright, Claude Code